AnalogOne

CLASS AB AUDIO AMPLIFIER

Discrete-Component Audio Power Amplifier

20 W

Class AB Audio Amplifier
Complementary MOSFET Output Stage
8 Ω Loudspeaker Application

Document Type	Technical Datasheet
Product	Class AB Audio Amplifier
Developer	Analog One
Institution	Tecnologico de Costa Rica
Technology	Discrete BJT and MOSFET

1. General Description

The Analog One Class AB Audio Amplifier is a discrete-component amplification system designed to increase audio signal amplitude and deliver power to an 8 Ω loudspeaker.

The amplifier consists of three functional stages: a BJT preamplifier, a MOSFET voltage amplification stage, and a complementary MOSFET Class AB power output stage.

The preamplifier performs initial signal conditioning and voltage amplification.

The intermediate stage provides the required voltage swing, while the Class AB output stage supplies the current necessary to drive the loudspeaker.

The voltage and power amplification stages operate with nominal symmetrical ±24 V power supplies.

The reference output power is 20 W.

A separate power-stage simulation recorded 22.8 W of average power delivered to an 8 Ω load under the specified test conditions.

2. Key Features

- Discrete-component audio amplifier.
- Three-stage amplification architecture.
- Common-emitter BJT preamplifier.
- MOSFET voltage amplification stage.
- Complementary Class AB output topology.
- Nominal ±24 V supply rails.
- 20 W reference output power.
- 8 Ω nominal loudspeaker load.

- Adjustable MOSFET bias.
- Source resistors for current stabilization.
- Output RC stabilization network.
- AC-coupled signal interfaces.

3. Functional Block Diagram

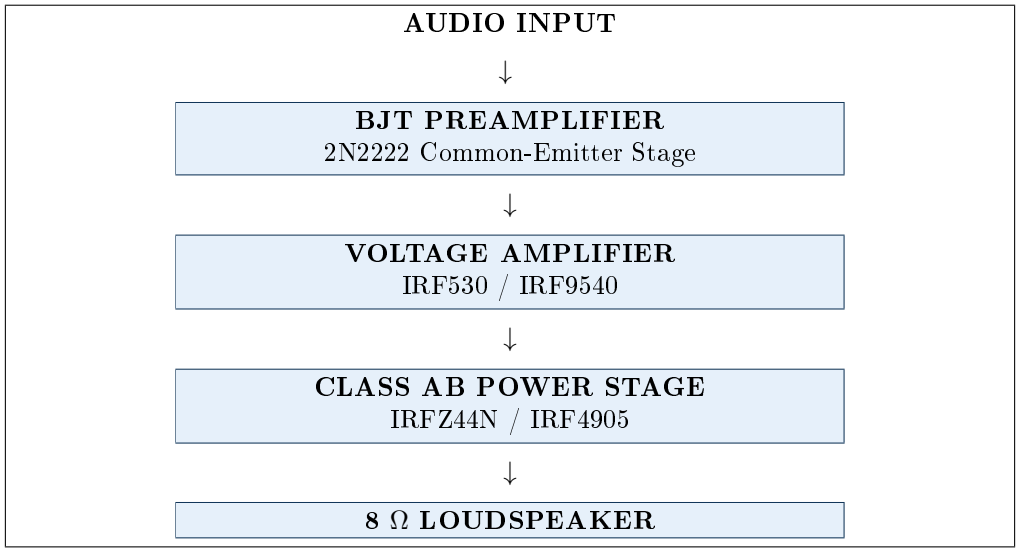


Figure 1: Functional architecture of the complete amplifier.

The input signal passes through the preamplifier, voltage amplification stage, and Class AB output stage before reaching the loudspeaker.

4. Electrical Specifications

Parameter	Value	Unit
Positive supply voltage	+24	V
Negative supply voltage	-24	V
Nominal load resistance	8	Ω
Reference output power	20	W
Simulated power-stage output	22.8	W
Reference RMS output voltage	12.65	V
Reference peak output voltage	17.89	V
Reference RMS output current	1.58	A
Reference peak output current	2.24	A

Table 1: Main electrical characteristics.

The reference electrical quantities correspond to the calculated requirements for a 20 W sinusoidal output into an 8 Ω load.

The simulated 22.8 W result corresponds to a separate power-stage test.

5. Preamplifier Stage

5.1 Description

The preamplifier uses a 2N2222 NPN bipolar junction transistor operating in a common-emitter configuration.

The emitter resistor provides negative feedback and stabilizes the operating point.

Input and output coupling capacitors block unwanted DC components.

5.2 Electrical Parameters

Parameter	Value	Unit
Transistor	2N2222	–
Supply voltage	24	V
Base voltage	7.39	V
Emitter voltage	6.69	V
Collector voltage	9.81	V
Quiescent collector current	3.02	mA
Collector-emitter voltage	3.12	V
Voltage gain	-2.13	V/V
Transistor power dissipation	9.42	mW
Estimated junction temperature	26.88	°C

Table 2: Calculated preamplifier characteristics.

6. Voltage Amplification Stage

6.1 Description

The voltage amplification stage consists of two IRF530 N-channel MOSFETs and one IRF9540 P-channel MOSFET.

The IRF530 transistors provide voltage amplification.

The IRF9540 operates as an active load.

The stage is designed to produce approximately 20 V peak output from a 2 V RMS input signal.

6.2 Electrical Characteristics

Parameter	Value	Unit
Supply voltage	±24	V
Input test voltage	2	V RMS
Target output voltage	20	V peak
Target voltage gain	7.07	V/V
Simulated gain at 1 kHz	7.118	V/V
Simulated gain at 20 Hz	6.930	V/V
Output fundamental at 1 kHz	20.112	V peak
Total harmonic distortion	1.183	%
Test load resistance	43	kΩ

Table 3: Voltage amplification characteristics.

The total harmonic distortion value corresponds to the individual voltage amplification stage, measured through the ninth harmonic.

7. Class AB Power Stage

7.1 Description

The power output stage uses a complementary MOSFET arrangement.

The IRFZ44N N-channel MOSFET and IRF4905 P-channel MOSFET reproduce the positive and negative portions of the amplified audio waveform.

The Class AB configuration reduces crossover distortion while maintaining moderate quiescent power consumption.

Adjustable bias networks establish the transistor operating conditions.

7.2 Output Characteristics

Parameter	Value	Unit
Nominal load resistance	8	Ω
Reference output power	20	W
Simulated load power	22.8	W
Simulated RMS output voltage	13.5	V
Simulated RMS output current	1.69	A
Peak-to-peak output voltage	38.2	V
Peak-to-peak output current	4.78	A
Simulation frequency	10	kHz

Table 4: Recorded Class AB power-stage results.

8. Power Supply Requirements

The voltage and power amplification stages operate from symmetrical DC supply rails.

Supply	Voltage	Unit
Positive rail	+24	V
Negative rail	-24	V
Reference ground	0	V

Table 5: Main supply requirements.

The preamplifier uses a nominal 24 V single-supply configuration.

All stages must share a suitable signal reference when interconnected.

9. Thermal Characteristics

The Class AB power stage dissipates energy through the output MOSFETs.

Adequate thermal management is necessary to maintain safe operating temperatures.

Device	Dissipation	Unit
IRFZ44N	6.58	W
IRF4905	6.63	W
Combined dissipation	13.21	W

Table 6: Recorded output-stage power dissipation.

These values correspond to the 10 kHz reference simulation.

Suitable heat sinks are recommended for the output MOSFETs.

Heat-sink selection must consider ambient temperature, junction temperature limits, and thermal resistance.

10. Component Summary

Stage	Component	Function
Preamplifier	2N2222	Common-emitter voltage amplification
Voltage stage	IRF530	N-channel voltage amplification
Voltage stage	IRF9540	P-channel active load
Power stage	IRFZ44N	Positive output current delivery
Power stage	IRF4905	Negative output current return
Power stage	0.22 Ω, 5 W	MOSFET source resistors
Power stage	100 Ω	MOSFET gate resistors
Power stage	10 kΩ	Adjustable bias trimmers
Power stage	4.7 μF	Signal coupling capacitors
Output network	10 Ω, 2 W	Zobel stabilization resistor
Output network	100 nF	Zobel stabilization capacitor
Load	8 Ω, 30 W	Nominal loudspeaker rating

Table 7: Main amplifier components.

11. Typical Application

The Analog One amplifier is intended for audio signal amplification using an external loudspeaker.

The typical operating sequence is:

1. Audio signal input.
2. BJT signal conditioning.
3. MOSFET voltage amplification.
4. Complementary Class AB current amplification.
5. Audio output to an 8 Ω loudspeaker.

The implementation requires proper supply connections, bias adjustment, grounding, and thermal management.

12. Operating Recommendations

- Verify power supply polarity before operation.
- Adjust the quiescent bias without an input signal.
- Verify the DC output voltage before connecting the loudspeaker.
- Use appropriate heat sinks for the output MOSFETs.
- Use correctly rated power resistors.
- Keep high-current output connections short.
- Verify the complete amplifier after interconnecting all three stages.

13. Performance Verification

The documented electrical results were obtained through circuit simulation.

The Class AB power-stage reference simulation used a 10 kHz input signal and an 8 Ω load.

The recorded average output power was 22.8 W.

The voltage amplification stage was separately evaluated using a 2 V RMS input signal at 1 kHz.

The simulated voltage gain was 7.118 V/V, with total harmonic distortion of 1.183%.

These results correspond to individual stage evaluations.

Further testing is necessary to determine the complete amplifier frequency response, harmonic distortion, noise, and continuous output-power rating.

14. Document Information

Developer	Analog One
Product	Class AB Audio Amplifier
Institution	Tecnologico de Costa Rica
Department	School of Electronics Engineering
Document	Technical Datasheet
Status	Design and Simulation

ANALOG ONE – AUDIO AMPLIFICATION SYSTEMS